COMPLEX ANALYSIS PHD EXAMINATION, AUGUST 2025

Answer SIX of the following eight questions. Explain your work in a precise and logical way. Any standard result may be used without proof but must be stated clearly and correctly.

Throughout the exam, $\mathbb{D}$ is the open unit disk in $\mathbb{C}$.

1. Let f be an analytic function on some open connected domain G and suppose that there exists an $a \in \mathbb{C}$ such that $f(z) = a$ has uncountably many solutions in G. What can be said about f?

2. State Morera's theorem and use it to prove that if $\{f_n\} \subset H(\mathbb{D})$ is a sequence of analytic functions that converges locally uniformly on $\mathbb{D}$ to a function f, then f is analytic.

3. Determine all entire functions f that satisfy $f(f(z)) = z$.

4. Let $p(z) = z^4 + 3z^3 - 1$. Is $1/p(z)$ analytic on the annulus $1 < |z| < 4$?

5. Evaluate the following improper integral:
$$\int_0^\infty \frac{1}{1+x^4}dx$$

6. Let f be a nonconstant entire function that is periodic with nonzero period $0 \neq T \in \mathbb{C}$. Prove that $f(z) - z = 0$ has infinitely many solutions.

7. Determine an explicit analytic bijection from $\mathbb{D} \cap \{z : \mathrm{im}(z) > 0\}$ to $\mathbb{D}$.

8. Let G be an open domain and let $\{f_n\} \subset H(G)$. Also let f be a nonconstant function such that $f_n \to f$ in $H(G)$. Let $a \in G$ and set $\alpha = f(a)$. Show that there is a sequence $\{a_n\} \subset G$ such that $a_n \to a$ and $f_n(a_n) = \alpha$ for all n sufficiently large.